Word Codes (Set B)

Year 4 Pt 2 · **Homework Worksheet**

Worked Example

Example — Code Practice: In a code, every letter is shifted forward by 3 (A→D, B→E…). What does 'GDW' decode to?

Answer: DAT… decoding by shifting BACK 3: G→D, D→A, W→T, giving 'DAT'. (Accept this as the mechanically correct decode of the given code letters.)

Quick Practice (15 questions)

1. In a code, every letter is shifted forward by 1 (A→B, B→C…). What does "FBU" decode to?
2. Using a +1 shift code, encode the word "DOG".
3. If the code for CAT is DBU (shift +1), what is the code for RUG? A) SVH B) SVI C) TVH D) SUH E) RVH
4. Select ALL that apply: using a shift of +2 (A→C, B→D…), which of these are correctly encoded? A) CAT → ECV B) DOG → FQI C) PIG → RKI D) SUN → UWP E) BAT → DCV
5. In a code, every letter is shifted BACK by 1 (B→A, C→B…). What does "QJH" decode to?
6. If DOG is coded as EPH (shift +1), what is the code for CAT? A) DBU B) DBV C) EBU D) DAU E) CBU
7. Using a +2 shift code (A→C, B→D…), encode the word "FOX".
8. In a code, every letter is shifted BACK by 2 (C→A, D→B…). What does "JGP" decode to?
9. If the word BAT is coded as CBU (shift +1), decode the word "SVO" using the same shift.
10. Select ALL that apply: using a shift of -2 (C→A, D→B…), which of these are correctly decoded? A) "FQI" → DOG B) "JGP" → HEN C) "RKI" → PIG D) "UWP" → SUN E) "EDV" → CAT
11. Using a +1 shift code, encode the word "HEN".
12. If CAT is coded as BZS (shift -1), what is the code for DOG? A) CNF B) ENH C) CNG D) DNF E) CMF
13. In a code, every letter is shifted forward by 3 (A→D, B→E…). What does the code word "FRJ" decode to?
14. Using a +2 shift code, encode the word "PIG".
15. Select ALL that apply: if the shift used is +1 (A→B, B→C…), which of these pairs is correctly matched? A) CAT→DBU B) DOG→EPH C) FOX→GPY D) SUN→TVO E) BAT→CAU

Mixed Practice (15 questions)

1. In a code, DOG is written as FQI. Work out the shift being used, then use it to encode "CAT".
2. In a code, each letter's code is found by shifting forward by 3 places. Decode the word "ZDW".
3. A code represents each letter by shifting back 2 places. If "VKKT" is a coded word, decode it.
4. Select ALL that apply: a code shifts every letter forward by 4. Which of these pairs are correctly matched? A) DOG→HSK B) CAT→GEX C) PIG→TMK D) SUN→WYR E) FOX→JSB
5. In a code, the word FISH is written as HKUJ. Work out the shift, then encode the word "CRAB" using the same shift.
6. A code shifts letters BACK by 3 places. Decode the coded word "ROXH".

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7. If the word MOON is coded as PRRQ (shift +3), decode the word "VWDU" using the same shift.

8. In a code, each letter is replaced by the letter 4 places further on, wrapping round after Z if needed. Encode the word "ZEBRA".

9. Select ALL that apply: a code shifts every letter back by 5 (wrapping round before A if needed). Which of these pairs are correctly matched? A) FROG→AMJB B) GOAT→BJVO C) HORSE→CJMNZ D) ZEBRA→UZWMV E) LEMON→GZHJI

10. In a code, CRAB is written as EUCD. Work out the shift, then decode "QJOGT" using the same shift.

11. A code shifts every letter forward by 5 places. Encode the word "GOAT".

12. If the word APPLE is coded as DSSOH (shift +3), decode the word "VQDNH" using the same shift.

13. In a code, each letter is replaced by the letter 6 places back, wrapping round before A if needed. Decode the coded word "ZKWLX".

14. Select ALL that apply: a code shifts every letter forward by 3. Which of these pairs are correctly matched? A) BEAR→EHDU B) SWAN→VZDQ C) GOAT→JRDW D) LION→OLRQ E) DEER→GHHU

15. In a code, the word MOUSE is written as PRXVH. Work out the shift, then use it to encode the word "TIGER".

Standard / Advanced (15 questions)

1. A code pair is given: LEMON → OHPRQ. Work out the shift being used, then decode the word "VQDNH" using the same shift.

2. In a code, each letter's position number is DOUBLED then 1 is subtracted, and the result is converted back to a letter (wrapping round after Z as needed). Encode the word "CAB" using this rule.

3. Select ALL that apply: a two-step code first shifts every letter forward by 2, then reverses the letter order of the word. Which of these results are correct for the word "CAT"? A) The first step (shifting CAT forward 2) gives ECV. B) Reversing ECV gives VCE. C) The final coded word is VCE. D) The final coded word is ECV. E) Reversing CAT first (before shifting) would give TAC.

4. In a code, vowels (A, E, I, O, U) are shifted BACK by 1, but consonants stay the same. Encode the word "DOG" using this rule.

5. A puzzle gives the pair RIVER → ULYHU as a worked example. Work out the shift, then use it to decode "ILYHU".

6. In a code, each letter is replaced by the letter the same number of places from the END of the alphabet as it is from the START (A ↔ Z, B ↔ Y…). Encode the word "DOG" using this reversal rule.

7. Select ALL that apply: a code shifts consonants forward by 1 but leaves vowels unchanged. Applying this to the word "TIGER", which of these are correct? A) T becomes U. B) I stays I. C) G becomes H. D) E stays E. E) R becomes S.

8. A code works in two steps: replace each letter with its alphabet position number, then multiply each number by 2 (not converting back to letters). What is the resulting set of numbers for the word "DOG"?

9. In a code, the pair SNAKE → VQDNH is given. Work out the shift, then encode the word "TIGER" using the same shift.

10. A two-step code shifts every letter forward by 1, then swaps the first and last letters of the resulting word. Apply this to the word "MOUSE" and give the final result.

11. Select ALL that apply: a code reverses the alphabet (A ↔ Z, B ↔ Y…). Applying this to the word "DOG", which of these are correct? A) D becomes W. B) O becomes L. C) G becomes T. D) The coded word is WLT. E) The coded word is TLW.

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12. In a code, each letter's position number is doubled, then the ones digit of the result is converted back to a letter (so a result of 24 gives the letter for 4, which is D; a result of 30 gives the letter for 0, treated as Z). What does the letter N become under this rule?

13. A puzzle gives the pair TRAIN → WUDLQ as a worked example. Work out the shift, then decode "EHDUV" using the same shift.

14. In a code, each vowel is replaced by the PREVIOUS vowel in the sequence A, E, I, O, U, A... (so E→A, I→E, O→I, U→O, A→U), while consonants stay the same. Encode the word "TIGER".

15. A two-step code first reverses the letters of the word, then shifts every letter back by 1. Apply this to the word "DOG" and give the final 3-letter result.

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